

4(a)	oscillations (of particles) are parallel to (the direction of) energy transfer	B1
4(b)(i)	(frequency varies as) vehicle moves relative to (stationary) observer	C1
	(vehicle) moving towards (observer) gives higher (observed) frequency (than 1.2 kHz) <u>and</u> (vehicle) moving away (from observer) gives lower (observed) frequency (than 1.2 kHz)	A1
4(b)(ii)	Doppler effect	B1
4(b)(iii)	position of vehicle labelled 'X' at top (12 o'clock) position on track	B1
4(b)(iv)	position of vehicle labelled 'Y' at right-hand edge (3 o'clock) position on track	B1
4(c)	maximum frequency = 1.40 (kHz) or 1.40×10^3 (Hz)	C1
	$1.40 = (1.2 \times 320) / (320 - v)$	C1
	$v = 46 \text{ m s}^{-1}$	A1
	or	
	minimum frequency = 1.05 (kHz) or 1.05×10^3 (Hz)	(C1)
	$1.05 = (1.2 \times 320) / (320 + v)$	(C1)
	$v = 46 \text{ m s}^{-1}$	(A1)